

A Guide for Implementing *DPYD* Genotyping for Systemic Fluoropyrimidines into Clinical Practice

Teresa T. Ho^{1,†} , D. Max Smith^{2,3,†} , Christina L. Aquilante^{4,5} , Emily J. Cicali⁶ , Nihal El Rouby^{7,8} , Daniel L. Hertz⁹ , Iman Imanirad¹⁰ , Jai N. Patel^{11,12} , Stuart A. Scott¹³ , Sandra M. Swain^{3,14} , Sony Tuteja¹⁵ , J. Kevin Hicks^{1,*}  and on behalf of the Pharmacogenomics Global Research Network Publication Committee

The safety of systemic fluoropyrimidines (e.g., 5-fluorouracil, capecitabine) is impacted by germline genetic variants in *DPYD*, which encodes the dihydropyrimidine dehydrogenase (DPD) enzyme that functions as the rate-limiting step in the catabolism of this drug class. Genetic testing to identify those with DPD deficiency can help mitigate the risk of severe and life-threatening fluoropyrimidine-induced toxicities. Globally, the integration of *DPYD* genetic testing into patient care has varied greatly, ranging from being required as the standard of care in some countries to limited clinical use in others. Thus, implementation strategies have evolved differently across health systems and countries. The primary objective of this tutorial is to provide practical considerations and best practice recommendations for the implementation of *DPYD*-guided systemic fluoropyrimidine dosing. We adapted the Exploration, Preparation, Implementation, and Sustainment (EPIS) framework to cover topics including the clinical evidence supporting *DPYD* genotyping to guide fluoropyrimidine therapy, regulatory guidance for *DPYD* genotyping, key stakeholder engagement, logistics for *DPYD* genotyping, development of point-of-care clinical decision support tools, and considerations for the creation of sustainable and scalable *DPYD* genotype-integrated workflows. This guide also describes approaches to counseling patients about *DPYD* testing and result disclosure, along with examples of patient and provider educational resources. Together, *DPYD* testing and clinical practice integration aim to promote safe prescribing of fluoropyrimidine therapy and decrease the risk of severe and life-threatening fluoropyrimidine toxicities.

The fluoropyrimidines, 5-fluorouracil and capecitabine, have been a fundamental part of chemotherapy regimens to treat solid tumors since their approval in the United States (US) over half a century ago.^{1,2} Numerous cancer types are treated with systemic fluoropyrimidines, including colorectal, pancreatic, breast, bladder, and head and neck cancers.^{2–4} For certain cancer types such as colorectal and pancreatic, fluoropyrimidines are typically part of frontline therapy. Each year over 2,000,000 patients worldwide are estimated to receive a fluoropyrimidine to treat their cancer.⁵

Fluoropyrimidines are highly effective anticancer agents that interfere with both DNA and RNA synthesis, resulting in DNA damage and inhibition of cell growth and proliferation.⁶ Despite

their efficacy, fluoropyrimidines are associated with significant toxicities, including gastrointestinal (e.g., vomiting, diarrhea, and mucositis), hematologic (e.g., neutropenia), and hand-foot syndrome (i.e., pain, erythema, and desquamation of the palms and soles). Importantly, the risk of severe fluoropyrimidine-associated toxicities is significantly higher among patients who have dihydropyrimidine dehydrogenase (DPD) deficiency,^{7–10} which is the enzyme responsible for 80% of fluoropyrimidine metabolism.^{11–13}

Deleterious germline variants in the *DPYD* gene are a primary cause of DPD deficiency, ranging from partial deficiency (reduced DPD enzyme activity) to complete deficiency (absent DPD enzyme

¹Department of Pathology, H. Lee Moffitt Cancer Center and Research Institute, Tampa, Florida, USA; ²Department of Oncology, Georgetown University Medical Center, Washington, District of Columbia, USA; ³MedStar Health, Columbia, Maryland, USA; ⁴Department of Pharmaceutical Sciences, University of Colorado Skaggs School of Pharmacy and Pharmaceutical Sciences, Aurora, Colorado, USA; ⁵Colorado Center for Personalized Medicine, University of Colorado Anschutz Medical Campus, Aurora, Colorado, USA; ⁶Department of Pharmacotherapy and Translational Research and Center for Pharmacogenomics and Precision Medicine, College of Pharmacy, University of Florida, Gainesville, Florida, USA; ⁷Department of Pharmacy Practice and Administrative Sciences, James L. Winkle College of Pharmacy, University of Cincinnati, Cincinnati, Ohio, USA; ⁸St Elizabeth Health Care, Edgewood, Kentucky, USA; ⁹Department of Clinical Pharmacy, University of Michigan College of Pharmacy, Ann Arbor, Michigan, USA; ¹⁰Department of Gastrointestinal Oncology, H. Lee Moffitt Cancer Center and Research Institute, Tampa, Florida, USA; ¹¹Division of Cancer Pharmacology & Pharmacogenomics, Atrium Health Levine Cancer Institute, Charlotte, North Carolina, USA; ¹²Atrium Health Wake Forest Baptist Comprehensive Cancer Center, Winston-Salem, North Carolina, USA; ¹³Department of Pathology, Stanford University, Stanford, California, USA; ¹⁴Georgetown Lombardi Comprehensive Cancer Center, Washington, District of Columbia, USA; ¹⁵Division of Translational Medicine and Human Genetics, Department of Medicine, Perelman School of Medicine, Philadelphia, Pennsylvania, USA. *Correspondence: (james.hicks@moffitt.org)

[†]Teresa T. Ho and D. Max Smith contributed equally to this manuscript as co-first authors.

Received October 21, 2024; accepted December 17, 2024. doi:10.1002/cpt.3567

activity). There is growing evidence indicating that pretreatment *DPYD* genotyping can mitigate the risk of severe and lethal fluoropyrimidine toxicity. *DPYD* genotyping and/or DPD phenotyping to guide fluoropyrimidine therapy is emerging as standard of care in many European countries, including direct regulatory guidance.¹⁴ In other regions such as the US, adoption of *DPYD* genotyping has been hampered by ongoing deliberations regarding the clinical utility of *DPYD*-guided fluoropyrimidine therapy, along with limited regulatory guidance and a lack of clinical practice guideline recommendations for *DPYD* genotyping.^{15,16} Nevertheless, several early adopters worldwide have incorporated *DPYD*-guided fluoropyrimidine dosing into clinical practice, demonstrating its feasibility.^{17,18} *DPYD* testing as part of a preemptive multi-pharmacogene panel has also been demonstrated to be feasible.¹⁹ The primary objective of this tutorial is to provide practical considerations and best practice recommendations for the implementation of *DPYD*-guided fluoropyrimidine dosing. Pharmacogenetics implementation can be guided by numerous frameworks and models.^{20,21} The Exploration, Preparation, Implementation, and Sustainment (EPIS) framework was adapted to describe suggested best practice considerations throughout the continuum of implementing *DPYD*-guided fluoropyrimidine dosing into practice (Figure 1), though the implementation science has not been conducted to systematically determine optimal frameworks or models. The association between *DPYD* variants and topical fluoropyrimidine toxicities has not been fully

elucidated, thus this tutorial is not applicable to topical fluoropyrimidine formulations.²²

Implementation of *DPYD*-guided fluoropyrimidine dosing is important globally as evidenced by *DPYD* testing recommended in several European countries. However, certain aspects of *DPYD* implementation (e.g., regulatory guidance, costs, and informatics infrastructure) can vary widely across regions and countries, and thus would be unwieldy to address for every region globally. This tutorial is applicable globally with the US health system used as an example to provide granular clinical and regulatory scenarios.

EXPLORATION PHASE

For health systems and clinicians considering implementation of *DPYD* genotype-guided fluoropyrimidine dosing, the goal of the exploration phase is to decide whether to implement it. A key first component is an understanding of foundational knowledge, including the evidence supporting implementation, available clinical guidelines, and the landscape of regulatory guidance.²³ However, the decision of whether to implement *DPYD* testing depends on considerations beyond published literature. Factors influencing stakeholder engagement (e.g., leadership goals, available resources, process barriers) will play a significant role in deciding whether to implement *DPYD*-guided fluoropyrimidine dosing. Herein, we briefly review concepts in the exploration phase including evidence supporting

Exploration	 Understand evidence supporting <i>DPYD</i> genotype-guided systemic fluoropyrimidines	 Engage multidisciplinary institutional champions and stakeholders	 Determine whether to implement <i>DPYD</i> testing for fluoropyrimidines in practice	
Preparation	 Determine logistics for <i>DPYD</i> genotyping	 Develop a system-level infrastructure to support <i>DPYD</i> testing	 Establish a clinician-level service to support implementation	 Create patient education material
Implementation	 Execute <i>DPYD</i> implementation plan	 Early monitoring of operational outcomes and processes		
Sustainment	 Create continuing education opportunities	 Maintain and/or refine CDS tools	 Obtain feedback from stakeholders and personnel	 Systematically evaluate clinical and operational metrics and communicate to key stakeholders

Figure 1 EPIS framework adapted for implementation of *DPYD*-guided fluoropyrimidine dosing.

the implementation of *DPYD*-guided fluoropyrimidine dosing and approaches to engaging key stakeholders.

Evidence supporting safety and efficacy of *DPYD*-guided fluoropyrimidines

Heterozygous *DPYD* carriers (i.e., intermediate metabolizers) have ~30–50% reduction in DPD activity and those homozygous for *DPYD* variants (i.e., poor metabolizers) are predicted to have 75–100% reduction in DPD activity depending on the specific *DPYD* variants present.^{8,24–26} Subsequently, 5-fluorouracil clearance is reduced by 40–80% among *DPYD* intermediate metabolizers compared with normal metabolizers.^{27,28} Meta-analyses involving thousands of patients have demonstrated a clear association between loss-of-function germline *DPYD* variants and severe fluoropyrimidine-induced toxicities.^{7,29} The incidence of severe toxicity (grade 3 or higher) among patients with DPD deficiency receiving standard doses of fluoropyrimidines is estimated to be 70–80%.^{9,30,31}

Prospective studies have demonstrated that preemptively performing dose reductions for those harboring deleterious *DPYD* variants can normalize systemic exposure to fluoropyrimidines and mitigate the risk of severe toxicities.^{32–35} Henricks *et al.* prospectively assessed the impact of *DPYD*-guided fluoropyrimidine therapy on the occurrence of severe toxicity among 1,103 evaluable patients.³⁵ For *DPYD**2A (c.1905+1G>A) carriers receiving a preemptive 50% fluoropyrimidine dose reduction, the frequency of severe toxicities was 31% compared with a historical control of 72%.³⁵ Recent studies also suggest *DPYD*-guided fluoropyrimidine dosing has similar efficacy compared with standard dosing.^{32,34,36} In a study with 931 evaluable patients, Knikman *et al.* demonstrated that *DPYD*-guided fluoropyrimidine dosing did not negatively affect overall survival, though progression-free survival data suggested a possible reduction in efficacy in patients with certain *DPYD* decreased function alleles, supporting the importance of titrated dose escalations if the initial dose reduction is tolerated.³⁶ Limitations of these studies include enrichment for colorectal and breast cancer patients, numerous different fluoropyrimidine regimens used, and controls consisting of either historical cohorts or comparisons to *DPYD* normal metabolizers rather than those with *DPYD* variants receiving full standard doses.

Clinical guidelines for *DPYD*-guided fluoropyrimidine dosing

DPYD-guided fluoropyrimidine dosing guidelines are available including the Clinical Pharmacogenetics Implementation Consortium (CPIC) and Dutch Pharmacogenetics Working Group (DPWG).^{37,38} A recent survey of US sites that have implemented *DPYD* testing showed that CPIC and DPWG were the preferred guidelines for recommendations on how to adjust fluoropyrimidine therapy based on *DPYD* results.¹⁷ Similarly, a survey developed by the Pharmacogenomics Global Research Network (PGRN) Implementation Working Group investigating *DPYD* implementation strategies found that globally CPIC and DPWG were the preferred guidelines for *DPYD*-guided fluoropyrimidine dosing.¹⁸

To promote standardized reporting of DPD activity, the CPIC guideline utilizes *DPYD* activity score to determine the predicted DPD phenotype.³⁷ Briefly, activity values of 0, 0.5, or 1 are assigned to each *DPYD* allele to define no function, decreased function, and

normal function, respectively. The sum of the activity value of each *DPYD* allele (e.g., *DPYD* c.1905+1G>A) making up the *DPYD* genotype (e.g., reference/c.1905+1G>A) yields an activity score and predicted DPD phenotype: normal, intermediate, or poor metabolizer (Figure 2). The DPD phenotype and activity score are then mapped to specific fluoropyrimidine dosing recommendations. Standard of care dosing is recommended for *DPYD* normal metabolizers. For patients identified as *DPYD* intermediate metabolizers, the published 2017 CPIC guideline recommended a 25–50% reduction of the starting fluoropyrimidine dose-dependent on the activity score.³⁷ However, in November 2018, an update was released on the CPIC website stating all *DPYD* intermediate metabolizers independent of activity score should be considered for 50% reduction of the starting fluoropyrimidine dose based on the study by Henricks *et al.*^{35,39} An important, often overlooked, CPIC recommendation for *DPYD* intermediate metabolizers is consideration for fluoropyrimidine-titrated dose escalation if the first two fluoropyrimidine cycles are tolerated. For *DPYD*-poor metabolizers, fluoropyrimidines should be avoided. For select *DPYD*-poor metabolizers who are not candidates for alternative agents, a strongly reduced fluoropyrimidine dose (e.g., >75% dose reduction) may be a consideration. The DPWG guidelines have similar recommendations to CPIC.³⁸ Other international groups have also developed guidance for *DPYD*-guided fluoropyrimidine dosing (Table 1).⁴⁰

To date, the National Comprehensive Cancer Network (NCCN) guidelines for breast, rectal, pancreatic, and head and neck cancer do not comment on *DPYD* genotyping.^{2–4,41} The NCCN colon cancer guideline (version 5.2024) acknowledges that *DPYD*-guided fluoropyrimidine dosing is feasible and has been shown to reduce the risk of severe toxicities, but the guideline panel did not recommend universal pretreatment *DPYD* genotyping.² The lack of a recommendation for routine *DPYD* genotyping was based on concerns for (1) risk of inferior efficacy with *DPYD*-guided dosing, particularly in the adjuvant setting, for certain *DPYD* variants, and (2) no CPIC dosing guidance that sufficiently comments on the different administration modalities including IV bolus 5-fluorouracil, infusional 5-fluorouracil, and oral capecitabine. These concerns are rational, but there is currently limited data demonstrating that standard of care fluoropyrimidine dosing has superior clinical outcomes compared with *DPYD*-guided fluoropyrimidine dosing to mitigate severe toxicity and death. Both the Henricks and Knikman studies included non-metastatic patients who were presumably receiving adjuvant fluoropyrimidine therapy, though outcome data specific to those with non-metastatic disease were not provided.^{36,42} The American Society of Clinical Oncology (ASCO) has not commented on *DPYD* genotyping.^{43–46} There have been recent calls for NCCN and ASCO to update their guidelines to include pretreatment *DPYD* testing for systemic fluoropyrimidine regimens, including from patient advocate groups such as the Advocates for Universal DPD/*DPYD* Testing.⁴⁷

Regulatory guidance for *DPYD* genotyping

The US Food and Drug Administration (FDA)-approved drug labels for fluoropyrimidines have long acknowledged the association between DPD deficiency and increased risk of serious adverse

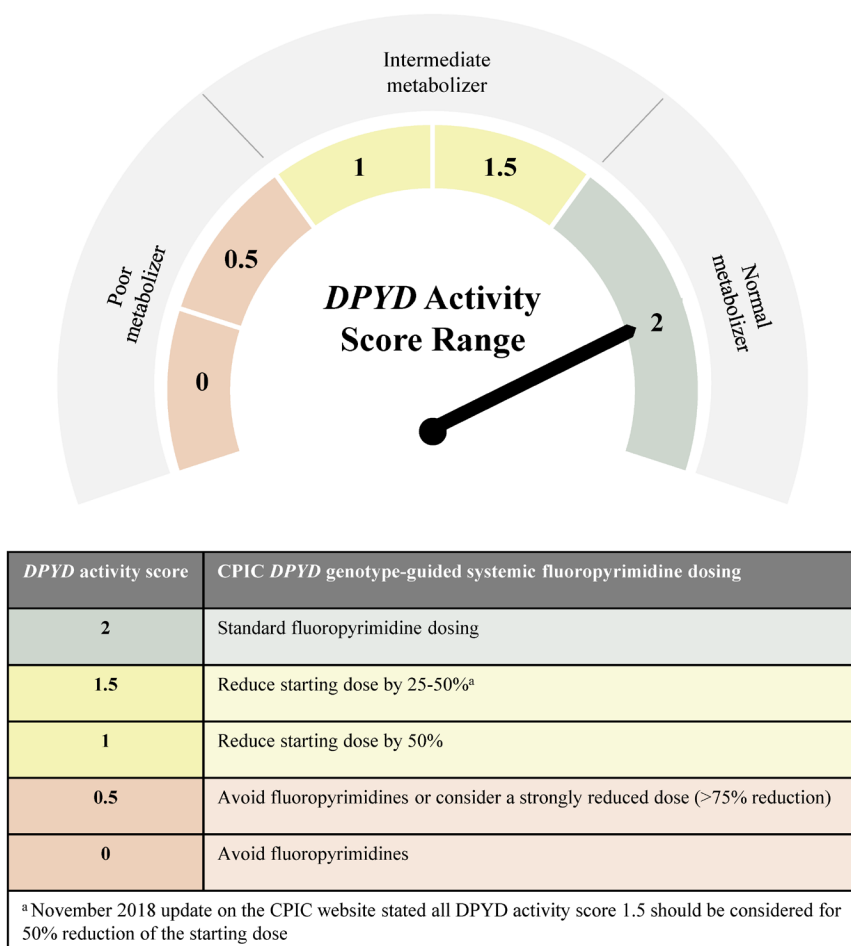


Figure 2 DPYD activity score range and CPIC dosing recommendations.

reactions, but historically have not included language that recommends *DPYD* testing. There have been calls for regulatory agencies in the US to incorporate pretreatment *DPYD* testing into guidelines for systemic fluoropyrimidine regimens including a Citizen's Petition to the FDA requesting a boxed warning for *DPYD* testing and fluoropyrimidine dose reductions in individuals with DPD deficiency.^{51,48,49} In response to the Citizen's Petition and as part of the Project Renewal initiative to update cancer drug labeling, the FDA-approved drug labels for 5-fluorouracil and capecitabine were recently amended to explicitly state that clinicians should "consider testing for genetic variants of *DPYD* prior to initiating [treatment]," and "inform patients of the potential for serious and life-threatening adverse reactions due to DPD deficiency and discuss with your patient whether they should be tested for genetic variants of *DPYD*," though a boxed warning or requirement for *DPYD* testing was not included in the labels.^{31,50,51} The FDA labeling provides limited guidance for dosing fluoropyrimidines in those who carry deleterious *DPYD* alleles. It is stated that there is no proven safe dose for patients with complete DPD deficiency.^{50,51} However, in contrast to the CPIC and DPWG guidelines, the FDA labeling for fluoropyrimidines states that there is insufficient data to recommend a specific dose for DPD intermediate metabolizers.

Examples of international *DPYD* and fluoropyrimidine regulatory guidelines are summarized in [Table 2](#).⁵²

Addressing the debate about pretreatment *DPYD* testing

During the exploration phase, there is likely to be debate about the pros and cons of pretreatment *DPYD* testing. Based on experiences from early implementors, commonly perceived deterrents include low frequency of *DPYD* variants, lack of randomized controlled trials assessing *DPYD*-guided fluoropyrimidine dosing, *DPYD* test turnaround time, costs, and lack of recommendations from clinical practice guidelines. Acknowledging and addressing such concerns before engaging with key stakeholders may enhance support for *DPYD* implementation.

An argument against *DPYD* implementation is the perceived small percentage of patients that harbor deleterious *DPYD* variants in the general population. However, it is estimated that between 6 and 9% of the general population carries a heterozygous deleterious *DPYD* variant, which varies by ancestry.^{16,48} Based on the Pharmacogenetic Pyramid framework previously established by the FDA, the frequency of *DPYD* variants combined with the number of unique patients receiving a fluoropyrimidine each year would meet medical need criteria.³¹ There may also be a misconception

Table 1 Clinical guidance for *DPYD* and fluoropyrimidines³⁹

Clinical guidance	<i>DPYD</i> normal Metabolizer	<i>DPYD</i> intermediate metabolizer	<i>DPYD</i> poor metabolizer
CPIC, 2017	Standard dosing	Activity score of 1.5: Reduce starting dose by 25–50% ^a Activity score of 1: Reduce starting dose by 50%	Activity score of 0.5: Avoid fluoropyrimidines or if alternative agents are not an option, consider a strongly reduced dose (> 75% dose reduction) Activity score of 0: Avoid fluoropyrimidines
DPWG, 2020	Standard dosing	Reduce dose by 50%	Avoid fluoropyrimidines
RNPGx, 2017	Standard dosing	Reduce dose by 50%	Avoid fluoropyrimidines
SEFF and SEOM, 2022	Standard dosing	Reduce dose by 50%	Avoid fluoropyrimidines
AIOM and SIF, 2019 ^b	100% of normal dose	75% of the normal dose	Avoid fluoropyrimidines
	c.1236GG (rs56038477)	c.1236GA (rs56038477)	c.1679GG (rs55886062)
	c.1679TT (rs55886062)		c.1905+1AA (rs3918290)
	c.1905+1GG (rs3918290)	50% of the normal dose	c.2846TT (rs67376798)
	c.2846AA (rs67376798)	c.1236AA (rs56038477)	
	c.2194GG (rs1801160)	c.1679TG (rs55886062)	
		c.1905+1GA (rs3918290)	
	85% of the normal dose c.2194GA (rs1801160)	c.2846AT (rs67376798)	
	70% of the normal dose c.2194AA (rs1801160)		

AIOM and SIF, Italian Association of Medical Oncology and the Italian Society of Pharmacology; CPIC, Clinical Pharmacogenetics Implementation Consortium; DPWG, Dutch Pharmacogenetics Working Group; RNPGx, Réseau national de pharmacogénétique (French National Network of Pharmacogenetics); SEFF and SEOM, Spanish Pharmacogenetics and Pharmacogenomics Society and the Spanish Society of Medical Oncology. ^aNovember 2018 update on the CPIC website stated all *DPYD* activity scores of 1.5 should be considered for a 50% reduction of the starting dose. ⁴⁰ ^bAIOM and SIF recommendations sorted based on CPIC assigned *DPYD* activity scores and phenotype.

Table 2 Regulatory guidance for *DPYD* and fluoropyrimidines^{51,52}

Regulatory agency	DPD deficiency testing recommendations	Dosing in intermediate metabolizers	Dosing in poor metabolizers
EMA	Phenotype and/or genotype testing prior to the initiation of treatment is recommended	A reduced starting dose should be considered to limit toxicity	Avoid fluoropyrimidines
FDA	Consider testing for genetic variants of <i>DPYD</i> prior to initiating treatment	Insufficient data to recommend a specific dose in patients with partial DPD deficiency	Avoid fluoropyrimidines
HCSC	Testing for DPD deficiency should be considered prior to treatment	Treat with extreme caution, initially with a substantial dose reduction and frequent subsequent monitoring and dose adjustment according to toxicity	Avoid fluoropyrimidines
PMDA ^a	Acknowledges association between DPD deficiency and toxicity	No specific recommendation	No specific recommendation
Swissmedic ^a	Testing for DPD deficiency recommended	Treat with extreme caution, initially with a substantial dose reduction and frequent subsequent monitoring and dose adjustment according to toxicity	Avoid fluoropyrimidines

EMA, European Medicines Agency; FDA, Food and Drug Administration; HCSC, Health Canada Sante Canada; PMDA, Pharmaceuticals and Medical Devices Agency of Japan. ^aTranslated from other sources per PharmGKB website.

that only *DPYD*-poor metabolizers, who account for less than 1% of the general population, are at risk of fluoropyrimidine-induced severe toxicity and death. Severe and lethal fluoropyrimidine-induced toxicities can also occur in *DPYD* intermediate metabolizers.²⁹

The main purpose of *DPYD* implementation is to identify patients at an elevated risk of toxicity and preemptively intervene to lessen that risk. What constitutes high-quality evidence supporting

an intervention is often debated, with randomized controlled trials considered the gold standard for clinical research. However, conducting randomized controlled trials to evaluate pharmacogenetic interventions are not always feasible.²³ For example, prospectively randomizing known *DPYD* carriers to receive either standard or reduced dosing is the most robust strategy to measure the impact of *DPYD*-guided fluoropyrimidine dosing. Such a study design would

be unethical to perform due to the undisputed risk of toxicity and death for those randomized to receive standard dosing. A prospective, randomized clinical trial assessing *DPYD*-guided vs. standard of care (no *DPYD* testing) would require an extremely large sample size that would likely be cost-prohibitive. *DPYD* studies utilizing historical controls for comparison groups, retrospective studies, and meta-analyses can provide sufficiently strong evidence to support an intervention and overcome the limitations of performing a prospective randomized controlled trial.

DPYD test turnaround time and costs can be major hurdles for implementation. Ideally, *DPYD* results should be returned before the initiation of a fluoropyrimidine, given the risk of early onset toxicity in *DPD*-deficient individuals. *DPYD* genotyping turnaround time is typically 5–12 days, depending on the laboratory used for testing.¹⁷ Many cancer centers, as part of performance metrics, have scheduling data that provide the average time from diagnosis or oncologist visits to receiving chemotherapy. It may be possible to determine the likelihood of *DPYD* results returned before a fluoropyrimidine is received by using estimated *DPYD* turnaround time and institutional data on time to receive chemotherapy. If it is estimated that most patients will receive chemotherapy before *DPYD* results are returned, implementation teams could consider meeting with clinicians and laboratory specialists to discuss clinical workflows and develop strategies for efficient return of *DPYD* results that can be applied in the preparation phase. This may include opportunities for moving *DPYD* ordering upstream in the workflow to account for reference laboratory processing times or performing in-house testing.^{17,53} Certain patients may need to start treatment immediately due to cancer-related symptoms, with workflows needed to accommodate clinical scenarios where a quick *DPYD* turnaround time is desired.

DPYD test costs depend on numerous factors including the laboratory used for *DPYD* testing, regional laws on biomarker test coverage, local insurance coverage policies, and private payer policies. *DPYD* genotyping is reimbursed by Medicare in most US states and a growing list of private payers reimburse for *DPYD* testing.^{31,54} Due to vast differences in *DPYD* genotyping costs and reimbursement, there is no one-size-fits-all approach. Before engaging key stakeholders, such as administrative leaders, implementation teams should consider researching local *DPYD* testing reimbursement policies and explore out-of-pocket costs, which typically are between 250 and 500 US dollars.³¹ If there is access to internal costs, a local budget impact analysis can be performed that takes into account estimated *DPYD* test costs and toxicity management including use of uridine triacetate, unplanned outpatient visits, emergency department visits, and hospitalizations.⁵⁵

Engaging key stakeholders

Stakeholders commonly include clinician champions, executive leadership, laboratory specialists, and informaticists. However, site-specific factors may lead to additional stakeholders (e.g., patient advocates and education specialists). A team of individuals with pharmacogenetics knowledge is often the driving force behind the important first step of engaging key stakeholders.^{56,57} However, a single individual with an interest in *DPYD* implementation is sufficient for engaging key stakeholders, which may be a

more common scenario in the community setting. For sites with limited pharmacogenetics expertise, clinical experts can be contacted for guidance and potentially on-site visits. Clinician champions and executive leadership are typically the first stakeholders to consider for engagement with site-specific factors determining which group to engage first.

Clinical champions who prescribe fluoropyrimidines can provide key insights into institutional standard of care practices and clinical workflows, while also encouraging the adoption of *DPYD* testing among colleagues. The audience for clinician champions in the exploration phase may be more targeted to key decision leaders rather than an all-inclusive group of clinicians. The extent of the audience may vary by institutional characteristics. For example, in the preparation phase, a multisite health system may invite providers who most commonly order fluoropyrimidines to gauge their interest in *DPYD* genotyping before expanding to additional clinicians.

In building a case for integrating *DPYD*-guided fluoropyrimidine therapy into practice, stakeholders need to understand the value proposition for testing. An educational session for clinician champions should include information on *DPYD* and fluoropyrimidines that highlights key studies demonstrating the clinical validity and utility of testing.^{56,58} A one-page summary (Figure S1) can be provided for clinician stakeholders for review. Consideration should be given to preemptively acknowledging potential barriers such as perceived low occurrence of *DPYD* variants, lack of randomized controlled trials, *DPYD* test turnaround time, and costs. Stakeholder educational sessions may also include data on the number of patients receiving fluoropyrimidines in the health system, frequency of fluoropyrimidine-related adverse events and hospitalizations, and frequency of uridine triacetate use from relevant institutional reporting systems.

Early support from executive leaders is also essential to maximize institutional buy-in, minimize potential roadblocks, understand desired key performance metrics, and garner dedicated resources (e.g., funding, additional personnel and informatics support) for the initiative. In addition to the clinical evidence supporting *DPYD*-guided fluoropyrimidine dosing, executive leaders may be interested in understanding how *DPYD* testing can reduce healthcare expenditures by avoiding the cost of unplanned patient visits and hospitalizations for fluoropyrimidine-associated toxicity along with the potential administration of the high-priced antidote, uridine triacetate.⁵⁹ Cost-effectiveness analyses indicate that *DPYD* screening is cost-effective, dependent on the genotyping costs, by increasing quality-adjusted life years.^{60–64} Furthermore, executive leadership may also have an interest in any associated liabilities. For example, a health system settled a wrongful death lawsuit after a patient died from capecitabine toxicity that included a 1-million-dollar payment to the patient's family and the requirement of oncologists to inform patients about *DPYD* testing.⁴⁸

Other key stakeholders to engage include institutional laboratory specialists and personnel for developing a *DPYD* assay. These individuals may also be involved when using an external laboratory. Clinical informaticists are essential for identifying electronic health record (EHR) limitations and processes for integrating *DPYD* results into the EHR in a discrete format so that automated clinical decision support (CDS) tools can be used to deliver

DPYD-guided fluoropyrimidine dosing recommendations at the point of care. Additionally, informaticists can help facilitate EHR-based test ordering.

One way to systematize the stakeholder engagement process is to embed qualitative methods (e.g., standardized questions) in early meetings, interviews, and/or focus groups. This strategy allows for a more structured approach to obtaining perspectives about barriers and facilitators of *DPYD* testing. This process also serves to ensure stakeholders' concerns are taken into consideration and preemptively addressed in the preparation phase. For example, one qualitative study highlighted stakeholder perceptions of key process barriers to implementing *DPYD* testing in practice, such as turnaround time, lack of integration of results into the EHR, and uncertainty regarding fluoropyrimidine dose modifications.⁶⁵ The study teams driving the implementation efforts were then able to design strategies to address the prominent barriers.^{66,67}

PREPARATION PHASE

When the decision is made to implement *DPYD* testing, institutions can transition to the preparation phase. A goal of this phase is to identify facilitators to overcome the barriers found in the exploratory phase, identify any new barriers along with solutions, and develop an implementation plan. Each individual institution will likely have a unique implementation plan based on local barriers and facilitators. However, based on experiences from early *DPYD* implementors all implementation plans will likely include (1) determining logistics for *DPYD* genotyping, (2) developing system-level infrastructure to support *DPYD* testing, (3) establishing clinician-level services to support the implementation process, and (4) creating patient education materials. Herein, we provide insights and best practice recommendations for each of these constructs.

Determining logistics for *DPYD* genotyping

Institutions must determine the target population for *DPYD* testing, which *DPYD* variants are critical to interrogate, and the preferred *DPYD* testing platform (e.g., targeted genotyping, full gene sequencing).

Which patients to test for *DPYD*. Ideally, any treatment-naïve patient who may receive a systemic fluoropyrimidine as part of their cancer treatment should be considered for *DPYD* testing. Patients who have tolerated a full-dose fluoropyrimidine over a period of several months are not likely to benefit from *DPYD* genotyping. However, it may be challenging to operationalize the omission of patients who have tolerated full-dose fluoropyrimidine therapy over an extended time, particularly when *DPYD* test ordering is driven by EHR-integrated tools. Overall, there are many institution-specific factors, including clinical workflows that may impact which patient populations are included in *DPYD* implementation efforts.

Certain patients may need to start a fluoropyrimidine regimen immediately due to cancer-related symptoms before *DPYD* results are available. *DPYD* results returned shortly after starting a fluoropyrimidine may be of clinical benefit in certain instances. For 5-FU infusions, *DPYD* results received within 48–72 hours after

initial administration may help identify those who would benefit from uridine triacetate, a pyrimidine analog that can lessen toxicity if given within the first 96 hours of fluoropyrimidine administration. Uridine triacetate is proposed to improve overall survival from 16% to 94% in those with severe fluoropyrimidine toxicity.⁶⁸ Case reports have also suggested that uridine triacetate may be of clinical benefit after the first 96 hours of fluoropyrimidine exposure, though further research is needed to clarify the ideal timing of uridine triacetate administration.⁶⁸ For oral capecitabine, dose adjustments can be made shortly after starting therapy to mitigate toxicities. Patients harboring a *DPYD* variant may also experience delayed severe fluoropyrimidine-related toxicity.⁶⁹ Thus, *DPYD* results returned after fluoropyrimidine therapy is initiated can assist with decision-making and side effect monitoring for subsequent cycles.

The clinical utility of *DPYD* genotyping for those experiencing fluoropyrimidine toxicities is debatable. If the fluoropyrimidine is planned to be discontinued and not restarted in the future, *DPYD* testing is less warranted. If fluoropyrimidine therapy is planned to continue, knowing *DPYD* metabolizer status can help guide dose adjustments or decision-making about switching to a different treatment regimen. For example, a *DPYD*-poor metabolizer with an activity score of 0.5 would require at least a 75% dose reduction. A clinician may not feel comfortable making such a drastic dose reduction without knowing *DPYD* results.

Another consideration is selecting a single clinic or malignancy (e.g., pancreatic cancer) to pilot *DPYD* implementation. An advantage of a pilot project is that it can help identify barriers to and facilitators of testing, and help refine strategies that can be used to inform expansion to other sites, clinics, and/or malignancies.⁵⁶ A pilot may also be suitable when clinician engagement is siloed by the patient population. For example, gastrointestinal oncology leadership may support *DPYD* implementation whereas others may not. Regardless, a successful pilot experience with engaged clinicians may convince others who were initially skeptical. A survey of US institutions that implemented pretreatment *DPYD* testing found that most sites (66.7%) implemented in select patient populations, highlighting that institution-specific factors greatly impact which patient populations are included in *DPYD* implementation efforts.¹⁷

Which *DPYD* variants to test. There is currently a lack of standardization across Clinical Laboratory Improvement Amendments (CLIA)-certified genetic testing in terms of which specific *DPYD* variants should be interrogated. Depending on the laboratory, the number of *DPYD* variants included in a test can range from three variants to full gene sequencing.¹⁷ The four main *DPYD* variants widely recognized for their established associations with fluoropyrimidine-induced toxicity include *DPYD**2A (c.1905+1G>A), *13 (c.1679T>G), c.2846A>T, and c.[1,129-5923C>G; 1236G>A]; however, please note that the star (*) allele system is no longer preferred when describing *DPYD* variants, as Human Genomic Variation Society (HGVS) nomenclature (<https://hgvs-nomenclature.org/>) is currently considered more appropriate.⁷⁰ To promote standardization of pharmacogenomic testing across clinical laboratories, the Association for Molecular Pathology (AMP) Pharmacogenomics

Working Group has recently published recommended *DPYD* variants for clinical testing, which is a key resource for selecting *DPYD* variants to test (Table 3).⁷⁰ *DPYD* allele frequencies differ across race and ethnicity, with both AMP and CPIC serving as resources for allele frequency data. Each institution's patient population inclusive of race and ethnicity should be taken into consideration when selecting which *DPYD* variants to test.

Selection of a *DPYD* genotyping test platform. Depending on institutional resources, *DPYD* genotyping may be performed through an in-house clinical laboratory or outsourced to an external laboratory, both of which must be CLIA-certified. Factors that may influence the selection of a genotyping platform include which *DPYD* variants are interrogated by the assay, desire for a single gene *DPYD* assay vs. a multigene panel, and preferred specimen(s) (e.g., blood, saliva, buccal swab). Guidance for selecting a laboratory for pharmacogenetic testing has previously been described.^{71,72} Potential advantages of in-house testing include quicker turnaround time, greater control over *DPYD* variants to include on a panel, interpretation, cost-control, and return of result workflows.⁵³ Potential advantages of using an external laboratory include less institutional resources needed and billing services, some of which may provide financial assistance plans. Somatic sequencing (i.e., tumor genome sequencing) is commonly performed in oncology, with some assays offering paired germline testing inclusive of *DPYD*.⁷³ Considerations for integrated germline and somatic testing have previously been described.^{73,74} Although such an approach may be an option in certain contexts,

not every patient may undergo somatic testing, or somatic testing may be performed after frontline therapy. Currently, the FDA is proposing to limit laboratory-developed tests (LDTs) and require FDA-clearance for all laboratory tests.³¹ Much is still unknown about this initiative and its potential impacts on *DPYD* testing. Currently, there is no FDA-cleared *DPYD* test.

Developing system-level infrastructure to support *DPYD* testing

The goal of developing system-level infrastructure to support *DPYD* genotyping is to optimize the delivery of healthcare by augmenting medical decisions with targeted clinical knowledge (e.g., *DPYD*-fluoropyrimidine dosing guidance) in the context of relevant patient information (e.g., *DPYD*-poor metabolizer phenotype, activity score 0.5).⁷⁵ This can be accomplished by seamless, end-to-end automated EHR integration spanning *DPYD* test ordering, discrete return of results, to clinical guidance at the point-of-care prescribing.^{67,76} Moreover, CDS alerts should be thoughtfully designed to increase effectiveness and avoid alert fatigue. The CPIC guideline for *DPYD* and fluoropyrimidines provides CDS resources to facilitate this process, including a general clinical workflow, *DPYD* result consult language and pre/post-test alerts, which can be further tailored to institutional preferences.³⁹ An example workflow is provided in Figure S2. A variety of CDS tools can support the ordering of *DPYD* testing and/or use of *DPYD* results in clinical decision-making.^{77,78} An automated process to support *DPYD* test ordering coupled with CDS support embedded within the EHR is an optimal approach, though the

Table 3 Association for molecular pathology guidance on *DPYD* variants for clinical testing⁶⁹

					Frequencies of <i>DPYD</i> alleles in biogeographical groups					
<i>DPYD</i> variants	Legacy name	rsID	DPD function	Activity value	African American	Central/ South Asian	East Asian	European	Latino	Sub-Saharan African
<i>TIER 1</i>										
c.1905+1G>A	*2A	rs3918290	No function	0	0.0031	0.0051	0.0	0.0079	0.00078	0.0
c.1679T>G	*13	rs55886062	No function	0	0.0	0.0	0.0	0.00056	0.0	0.0
c.1129-5923C>G, c.1236G>A	HapB3	rs75017182 rs56038477	Decreased function	0.5	0.0031	0.019	0.0	0.024	0.0059	0.0
c.557A>G	–	rs115232898	Decreased function	0.5	0.012	0.0	0.0	0.000088	0.0012	0.026
c.868A>G	–	rs146356975	Decreased function	0.5	0.0	0.0	0.0	0.0	0.00034	0.0020
c.2279C>T	–	rs112766203	Decreased function	0.5	0.0	0.0060	0.0	0.0	0.0	0.0
c.2846A>T		rs67376798	Decreased function	0.5	0.0031	0.00064	0.0	0.0037	0.0021	0.0
<i>TIER 2</i>										
c.299_302del	*7	rs72549309	No function	0	–	0.0	0.0	0.00018	0.00010	–
c.703C>T	*8	rs1801266	No function	0	–	0.00020	0.0	0.000091	0.0	–
c.1314T>G	–	rs186169810	Decreased function	0.5	0.0	0.0	0.00068	0.0	0.0	0.0
c.1475C>T	–	rs72549304	No function	0	–	0.00020	0.0	0.0	0.0	–
c.1774C>T	–	rs59086055	No function	0	0.0	0.0	0.0016	0.000013	0.0	0.0
c.2639G>T	–	rs55674432	No function	0	–	0.00070	0.0	0.0	0.0	–

development of system-level infrastructure will depend on each institution's EHR software capabilities and resources for building EHR tools.

DPYD test ordering. An ideal approach is integrating *DPYD* testing into order sets or treatment plans. For example, clinics may have an order set that includes all the necessary laboratory tests and imaging that need to be performed specific to a new cancer diagnosis. *DPYD* testing can also be integrated into treatment plans containing laboratory tests that must be completed before chemotherapy initiation. A potential pitfall with this approach is that depending on the EHR and institutional governance of oncology order sets and treatment plans, each provider can modify and then save laboratory orders that are unique to their practice. Thus, *DPYD* testing that is incorporated into core order sets or treatment plans may not show up on individualized plans. Pretest CDS alerts can also be triggered at the time a fluoropyrimidine is ordered to recommend *DPYD* testing, with the *DPYD* test orderable embedded in the CDS alert (Figure S3). Pretest alerts can be an opt-out design where the provider can decline to order *DPYD* testing, which was shown to increase the testing rate from 9 to 96% at Dana Farber Cancer Institute.⁶⁷ *DPYD* test implementation may differ across clinics. For instance, a pancreatic clinic may prefer to have *DPYD* testing automated via an order set, as pretest CDS alerts may be too disruptive given the expected frequency of alert firing. A breast clinic may prefer pretest CDS alerts at the time of ordering a fluoropyrimidine since this drug class is not used for every breast cancer patient and is often a later-line therapy. Rules within the EHR should be developed to prevent duplicate *DPYD* test ordering and prevent a pretest alert from firing if a *DPYD* test has already been ordered or if the provider previously declined to order *DPYD* genotyping.

If *DPYD* test ordering cannot be fully automated within the EHR, advanced practice professionals, pharmacists verifying chemotherapy orders, and nurses can help facilitate the test-ordering process as another safeguard mechanism and help enforce any *DPYD* testing policies a health system may create.^{56,79} Additionally, studies are evaluating the utility of precision medicine navigators in identifying patients eligible for genetic testing.⁸⁰

Return and storage of *DPYD* test results. The method by which *DPYD* test results are returned to clinicians and patients varies by laboratory. Laboratories may return pharmacogenetic results as a PDF (i.e., non-structured data) obtained through an external portal, or as structured (discrete) EHR data. Ideally, *DPYD* results (e.g., genotype, phenotype, and/or activity score) should be documented discretely in the EHR to drive post-test CDS.⁷⁷ A coordinated effort between clinical laboratory and health informatics personnel is needed to store discrete *DPYD* results, whether it be through tools to pull data from a PDF, obtaining results directly from the laboratory in a file that enables results to be embedded discretely in the EHR, or establishing an interface between the laboratory and EHR to communicate data between systems using standardized formats (e.g., HL7). If these approaches are not feasible, then *DPYD* result interpretations may need to be

entered manually into the EHR if the goal is to use post-test CDS tools in downstream clinical applications. There are examples of pharmacogenetic clinical services that review results and enter a consult note containing discrete phenotype results to drive EHR post-test CDS applications (Figures S4 and S5).⁸¹ EHR software may also have dedicated sections to organize genetic information that can be easily accessed, or the ability to create a section in the EHR (e.g., genomics tab) to organize genetic data.⁸² For results embedded discretely in the EHR, triggers for the CDS alerts should ideally be based on *DPYD* activity score to provide more granular recommendations. Alternatively, an institution may consider using the genotype or phenotype assignment as an alert trigger. Local laws and guidance for health and genetic information privacy should be considered when developing strategies for storing and disseminating genetic information.

DPYD test result interpretation. A prior survey by Tracksdorf *et al.* showed that pharmacists and medical oncologists are the primary clinicians responsible for *DPYD* test interpretation.¹⁷ For over 60% of survey respondents, clinician interpretation was supplemented with CDS. Decision support may take a variety of forms including interruptive post-test alerts, passive CDS such as a pharmacogenetics consultation note (Figure S5), or an in-basket message to providers.^{82,83} In crafting language for *DPYD*-fluoropyrimidine interpretative CDS tools, feedback from key stakeholders is important as some institutions require a vetting process for approval, such as an institutional Pharmacy & Therapeutics (P&T) committee.⁸³

Establishing clinician-level services to support the implementation process

A clinician-level support system for *DPYD* implementation is also recommended to ensure the adoption and integration of *DPYD* testing in practice. Larger medical centers may have a pharmacogenetics or precision medicine clinical service to support *DPYD* implementation. Other clinical departments, such as pharmacy, with individuals trained in pharmacogenetics can also help lead *DPYD* implementation. An important offering is a clinical consultation service to assist clinicians with questions about using *DPYD* results to guide fluoropyrimidine therapy. A pharmacogenetics team member and/or a trained oncology pharmacy specialist may be assigned the responsibility of creating a consult note in the EHR and directly notifying the clinical team about high-risk *DPYD* results.⁸⁴ Follow-up on *DPYD* results by dedicated personnel can be particularly helpful if a patient starts fluoropyrimidine-based treatment before results are available.⁷⁷

Other important deliverables include education for all relevant clinicians. Educational modalities may include grand rounds, in-services, online modules, among others. Education modalities that provide continuing education credit may promote participation. Education should include information about clinical evidence supporting *DPYD* genotyping, who should be considered for *DPYD* testing, test ordering and EHR tools, dosing recommendations, and patient education resources. It is strongly recommended to develop a short (e.g., 1 page) reference document for clinicians that briefly describes why *DPYD* testing should be performed, how to

order a *DPYD* test, *DPYD*-guided fluoropyrimidine recommendations, and any additional considerations or limitations (Figure S1). Ongoing *DPYD* education may be needed at certain intervals to capture new trainees or clinicians and provide updates to new clinical data or guideline recommendations. Targeted in-services, such as case-based discussions, can remind clinicians of *DPYD* testing workflows.⁸⁵ Point-of-care learning may also take place on a wider scale if electronic CDS tools are integrated into clinical workflows. Other education opportunities include short *DPYD* educational sessions at multidisciplinary tumor boards along with pharmacy and nursing departmental meetings. There are examples of health care organizations using online educational tools, such as Test2Learn for pharmacogenetics education.^{84,86} The Test2Learn educational tool can be customized to germline testing in oncology, inclusive of *DPYD*. Professional organizations such as the American Society of Hospital Pharmacists also provide pharmacogenetic educational tools, many of which are virtual.²³

Educating patients on *DPYD* testing

Recent updates to the FDA labeling of fluoropyrimidines have assigned the responsibility to the healthcare provider for informing patients about the toxicity associated with DPD deficiency and the availability of *DPYD* testing – underscoring the importance of patient education. A logical time to inform a patient about *DPYD* testing is when a provider recommends a fluoropyrimidine-based chemotherapy regimen for the patient's cancer. Given the time constraints of a provider visit, a more in-depth discussion about *DPYD* testing may be supplemented through the development of education materials or when a patient meets with support personnel (such as advanced practice provider, pharmacist, and/or nurse). Chemotherapy teaching visits with a nurse or pharmacist may also be an option for *DPYD* testing and education, depending on whether the education takes place before, during, or after chemotherapy administration. Telemedicine-based approaches may also be an option for patient education. A guide for modeling pretest pharmacogenomic counseling has previously been described, which can be adapted for *DPYD* counseling.⁸⁷ Specific counseling points to consider for *DPYD* testing include information on the purpose of testing, benefits and limitations of testing, risk mitigation strategies for DPD deficiency (i.e., dose reduction or switching therapy), type of sample required for testing, turnaround time, the antidote for fluoropyrimidine toxicity, and cost of testing. Patients should also be made aware that the absence of *DPYD* variants (i.e., *DPYD* normal metabolizer result) does not obviate the risk of fluoropyrimidine toxicity.

Various methods can be used to return results to a patient including verbal return of results over the phone or during a virtual or in-person clinic visit, results provided in the EHR patient portal, or electronically communicated with a copy of the results. Tracksdorf *et al.* reported that patient-specific portals were used by over 70% of respondents to return *DPYD* results.¹⁷ In addition, over 60% of respondents reported that the medical oncologist discusses *DPYD* results with the patient. A guide for modeling post-test pharmacogenetic counseling has previously been described and may be adapted for disclosing *DPYD* test results.⁸⁸ It is important for the patient to understand whether their *DPYD* result predisposes them to severe fluoropyrimidine toxicity and the need for

a modified chemotherapy regimen. Examples of *DPYD*-related patient education material can be found in Figure S6. These educational materials can supplement information given by clinicians during medical visits. Because medical oncologists may be the clinicians most often disclosing results with patients, it is important to work closely with oncologists for the development of educational materials. Digital tools disseminated to patients via the EHR portal, email, and cell phones are being explored to deliver genetic education, including *DPYD*, and may emerge as the preferred education method in the future.⁸⁹

A patient may be seen at a large cancer center for initial diagnosis or a second opinion, and then receive most of their care in community settings. Pharmacogenetic test results may not be communicated among the disparate health systems a patient is receiving care from. Thus, it is important to educate patients about sharing their *DPYD* results with all health care providers.

In addition to fluoropyrimidine toxicity, *DPYD* variants are also associated with Mendelian DPD deficiency, an autosomal recessive disorder of pyrimidine metabolism.⁸⁶ This condition has a variable presentation, ranging from no symptoms to severe neurological symptoms with motor and intellectual disabilities. Due to the varying penetrance, there is debate as to whether DPD deficiency alone would trigger symptoms.^{87,88} For patients who are carriers of *DPYD* variants, there might be a benefit from genetic counseling for family planning purposes. Furthermore, if there is a family history of cancer there may be benefit of *DPYD* cascade testing. For example, a patient diagnosed with Lynch syndrome and found to carry a *DPYD* variant would have blood relatives with an elevated risk of colorectal cancer and exposure to fluoropyrimidines. However, no established consensus exists on DPD deficiency genetic counseling practices or *DPYD* cascade testing.

IMPLEMENTATION PHASE

Once the preparation phase is complete and leadership has given approval, *DPYD* testing can be implemented into patient care. Multiple stakeholders play a role in *DPYD* implementation; thus, coordination of go-live activities is essential. Institutions should consider setting a date for go-live that occurs earlier in the week. For example, a go-live on a Friday may result in issues arising over the weekend where there may be limited support. Upon implementation, initial *DPYD* testing should be tracked closely for success stories, areas for improvement, and opportunities to adapt processes to better fit real-world situations and clinical workflows. Feedback from key stakeholders and personnel involved with *DPYD* testing is essential for early adaptation of *DPYD* implementation initiatives. Key performance metrics to consider tracking include *DPYD* testing rate, concordance with *DPYD* recommendations, and if applicable number of interruptive alerts fired and acceptance of alerts (Table 4).⁷⁸ Development of EHR dashboards or automated reports (e.g., *DPYD* tests ordered, new fluoropyrimidine orders) can assist with streamlining data collection during the implementation and sustainment phases.

SUSTAINMENT PHASE

The goal of the sustainment phase is to continue to utilize *DPYD*-guided fluoropyrimidine therapy while adapting to both lessons

Table 4 Implementation metrics to consider tracking

Metric type	Data tracking
DPYD testing	Number of DPYD tests ordered Turnaround time by tracking date test ordered, blood drawn, and result returned Percentage of DPYD results returned before fluoropyrimidine treatment initiation Frequencies of reported DPYD genotypes and phenotypes Reimbursement (if billing)
Treatment plan and outcome	Fluoropyrimidine treatment-naïve vs. experienced patients Treatment regimens and dosages Concordance of treatment with DPYD recommendations Occurrence of fluoropyrimidine-related adverse events Uridine triacetate use Fluoropyrimidine-related unplanned outpatient visits Emergency department visits Hospitalizations Death
CDS	Alerts fired User interactions with the alerts (e.g., accept or override) Number of clinical consults/notes Clinical actions taken in response to CDS tools

learned and updates to evidence such as clinical guidelines. In most instances, key performance metrics should continue to be tracked long-term during the sustainment phase. This can provide long-term information about the penetration and acceptance of DPYD implementation efforts. If rates of DPYD testing decrease over time or if there is a reduction of concordance with DPYD recommendations, this may signify the need to re-engage clinicians about the importance of DPYD genotyping. A mechanism for clinician continuing education will be needed to address evolving evidence for DPYD-guided fluoropyrimidine therapy and associated clinical and regulatory guidance. CDS tools may require maintenance and/or further refinement in response to new DPYD evidence, and clinician workflow changes and/or preferences. The DPYD testing platform should be reviewed to determine if there continues to be appropriate coverage of variants based on AMP guidance and evolving data. DPYD test volumes may change over time which could require reevaluation of resources needed for sustainment. For example, if DPYD testing increases by 50% due to the growth of the health system, then additional pharmacogenetic specialists and laboratory support may be needed for scalability. To streamline operational monitoring during the sustainment phase, an institution may consider additional EHR tools such as a dashboard or queue for monitoring the return of results and DPYD test orders.

OTHER PRACTICE CONSIDERATIONS FOR IMPLEMENTING DPYD TESTING

The implementation of any new evidence-based intervention, including DPYD testing, invariably will encounter challenges and obstacles. Inertia can present in situations where established traditions and routines impede the uptake of a new intervention. Gaps in the literature may leave clinicians perplexed about how to approach the dosing of fluoropyrimidines in certain clinical scenarios for a particular DPYD result. In addition to DPYD pharmacogenetics, other clinical variables can also influence fluoropyrimidine toxicity. To address these potential obstacles, we provide additional practical considerations for the DPYD implementation process.

Dosing approaches in different cancer types

Fluoropyrimidine-based therapy is used to treat numerous solid tumors, with dosage varying based on the cancer type and the regimen used including single-agent fluoropyrimidines or in combination with chemotherapy and/or radiation therapy. Both the Henricks and Knikman studies included numerous cancer types and diverse fluoropyrimidine regimens.^{35,36} However, these studies were not powered to compare outcomes between cancer types or regimens. Most studies to date have been enriched for colorectal cancer that tends to be treated with higher fluoropyrimidine doses. Extrapolation of those data to less studied tumor types, or instances where lower fluoropyrimidine doses may be used, can be challenging. Furthermore, it is unrealistic to examine optimal magnitudes of DPYD-based dose reductions for every possible dosing regimen. For patients started on a lower fluoropyrimidine dosage, providers may not want to further reduce the dose based on DPYD results. However, DPYD results can help guide titration and anticipated maximum tolerated dose. Furthermore, DPYD-poor metabolizers may not tolerate lower starting dosages, arguing that DPYD testing should be performed regardless of the planned starting dose or tumor type.

Optimal dose reduction

The optimal dose reduction for fluoropyrimidines among deleterious DPYD variant carriers has been an active area of research to mitigate adverse effects but maintain treatment efficacy. Of particular interest are patients who are DPYD intermediate metabolizers with an activity score of 1.5 (e.g., c.1236G>A and c.2846A>T carriers). The 2017 DPYD CPIC guideline recommended a 25–50% dose reduction for those with an activity score of 1.5, but later amended that recommendation to a 50% dose reduction based on the Henricks *et al.* study suggesting that a 50% dose reduction may be more appropriate for reducing the risk of fluoropyrimidine toxicity.^{35,37} Subsequently, Knikman *et al.* showed that DPYD-guided fluoropyrimidine therapy does not negatively affect progression-free survival or overall survival.³⁶ However, this study raised concerns

about the appropriate combination of dose reduction and titration for those with an activity score of 1.5. In the Knikman study, both *DPYD* c.1236G>A and c.2846A>T variant carriers received a 25% dose reduction, which is notably different from current CPIC recommendations.^{36,39} A greater percentage of *DPYD* c.1236G>A carriers (27.3% compared with 17.2% matched *DPYD* normal metabolizers) and especially *DPYD* c.2846A>T carriers (66.7% compared with 18.5% *DPYD* normal metabolizers) experienced grade 3 or higher toxicities. However, progression-free survival was significantly lower for *DPYD* c.1236G>A carriers and trended lower, but not significant, for *DPYD* c.2846A>T carriers. Only a small percentage of patients underwent fluoropyrimidine dose titration.

Taking observations from both the Henricks and Knikman studies together, a 25% fluoropyrimidine dose reduction may not be sufficient to prevent severe toxicities among *DPYD* c.2846A>T carriers.^{35,36} Additionally, both *DPYD* c.1236G>A and c.2846A>T variant carriers would benefit from fluoropyrimidine dose titration to improve progression-free survival, if the initial fluoropyrimidine dose is tolerated. Based only on the Henricks and Knikman studies, it is difficult to state what the optimal dose reduction is for *DPYD* c.1236G>A carriers and is out of scope for this tutorial to make a dosage recommendation.

DPYD c.1236G>A is part of the HapB3 haplotype that consists of both the benign c.1236G>A variant and the causal c.1129-5923C>G intronic splice variant associated with decreased DPD activity. Historically, reference laboratories used the c.1236G>A variant as a tag to detect the presence of the c.1129-5923C>G causal variant on the assumption of complete linkage disequilibrium (i.e., c.1129-5923C>G is present when c.1236G>A is detected) based on previous studies in Europeans—particularly for laboratories performing exome sequencing where intronic variants, such as c.1129-5923C>G, are not detected.^{37,90} However, recent data demonstrated that the two variants are not in complete linkage disequilibrium in all populations.⁹¹ As such, in very rare instances (estimated 0.2% of population) a patient may harbor the benign c.1236G>A variant but not the c.1129-5923C>G causal variant, and thus would incorrectly be predicted to be a *DPYD* intermediate metabolizer if only genotyping the tag marker. Genotyping capabilities have evolved over time and a growing number of reference laboratories directly test for *DPYD* c.1129-5923C>G, which is currently recommended by AMP.

AMP recently expanded the recommended *DPYD* variants for clinical testing, several of which are predominantly observed in understudied minority populations.⁷⁰ Additionally, as exome and whole genome sequencing expands clinicians may be exposed to *DPYD* variants with limited data describing effects on fluoropyrimidine metabolism. CPIC supplemental tables provide predicted activity scores for numerous *DPYD* variants, including those in AMP testing recommendations, and can be a guide for dose adjustments. A potential approach to rare *DPYD* variants with limited clinical data is an initial lower fluoropyrimidine dose coupled with close clinical monitoring and dose titration to balance severe toxicity mitigation and efficacy. Though, additional research is needed to further refine optimal fluoropyrimidine dose adjustments among rare *DPYD* variants. In certain instances, case reports may help to evaluate toxicity risk and dose optimization

for rare *DPYD* variants.⁹² The *DPYD* CPIC guideline is currently being updated and should be monitored for any changes to *DPYD*-guided dosage recommendations and any changes to allele activity score. Clarity regarding when and how to perform fluoropyrimidine titration in those harboring *DPYD* variants should also be addressed in future research and guideline updates.

Titration protocol considerations

Titration dose escalation in patients receiving *DPYD*-guided preemptive dose reductions is important to ensure each patient achieves their individualized maximum tolerated dose.⁴² There are limited studies evaluating the impact of fluoropyrimidine dose titration after initial *DPYD*-guided dose reduction. A recent study describing *DPYD* implementation used a 10% fluoropyrimidine dose increase in variant carriers who tolerated the first two cycles of therapy without toxicity, which was based on available literature and expert opinion.⁹³ In a study from the Netherlands, investigators evaluated tolerance-based capecitabine dose escalation after *DPYD*-guided dosing in variant carriers.⁹⁴ One hundred and eighty patients were included, 11 of whom were heterozygous carriers who received *DPYD*-guided dose reductions (~50%). The mean dose escalation was 8.5% with one patient experiencing a grade 3 toxicity after dose escalation. In the Henricks *et al.* study, 11 *DPYD* variant carriers had dose escalations of an unreported magnitude, 5 of whom were unable to tolerate the escalation, leading to subsequent dose reductions and treatment discontinuation.³⁵ Although the magnitude of dose increases was not reported, these data emphasize the need to be cautious with dose escalations. Based on these limited studies, it may be reasonable to consider a 5–10% increase in dose after tolerating the first two cycles of fluoropyrimidine chemotherapy among deleterious *DPYD* variant carriers undergoing a preemptive dose reduction. *DPYD* implementation strategies should include workflows to ensure eligible patients undergo dose titration. This may include leveraging dashboards or other reporting systems with a dedicated clinician (e.g., pharmacist) that follows up on patients eligible for fluoropyrimidine titration.

Other factors influencing fluoropyrimidine-associated toxicity

Variability in fluoropyrimidine toxicity is influenced by factors such as age, ancestry, sex, renal function, and body composition.^{95–97} A prior study developed a predictive model for capecitabine toxicity in *DPYD* normal metabolizers, including 1,745 patients with cancer treated with capecitabine-based regimens. Both univariable and multivariable logistic regression demonstrated that age, sex, and type of treatment regimen were strong predictors of capecitabine-related toxicity.⁹⁸ The development of risk tools that integrate *DPYD* status and other clinical factors may lead to more precise dosing. Such tools may be inclusive of physiologically based pharmacokinetic models.⁹⁹

DPD phenotyping

Due to the limitations of *DPYD* genotyping, it has been suggested that DPD phenotyping by measuring uracil concentrations may be a more accurate biomarker of DPD activity. However, clinical

DPD phenotyping is currently not available in the US and has its own limitations.¹⁰⁰ A study of 955 patients without *DPYD* variants identified large inter-center differences in pretreatment uracil levels among patients. One potential cause for the poor performance of pretreatment uracil levels as a DPD phenotyping tool is that improper pre-analytical storage conditions can result in uridine conversion into uracil, which would confound the analysis.¹⁰¹ It is possible that pre-analytical handling concerns can be addressed by the use of strict pre-analytical requirements, as performed as a standard of care in France since 2019.¹⁰¹ Of note, some countries perform both *DPYD* genotyping and DPD phenotyping.

CONCLUSION

In the pursuit of prescribing fluoropyrimidines safely, *DPYD* testing should be accessible to all patients and efficiently integrated into clinical practice. In this tutorial, the EPIS framework was adapted as a model for implementing *DPYD*-guided fluoropyrimidine dosing into patient care. The suggested best practice recommendations, practical considerations, and resources provided herein are intended to serve as a guide to support and promote the implementation *DPYD*-guided fluoropyrimidine dosing into clinical practice. Together, *DPYD* testing and clinical practice integration aim to promote safe prescribing of systemic fluoropyrimidine therapy and decrease the risk of severe and life-threatening fluoropyrimidine toxicities.

SUPPORTING INFORMATION

Supplementary information accompanies this paper on the *Clinical Pharmacology & Therapeutics* website (www.cpt-journal.com).

CONFLICTS OF INTEREST

DLH is an unpaid medical advisor for Advocates for Universal *DPYD*/DPD Testing and has received relevant research funding from LaCar MDX Technologies, the University of Michigan Office of the Vice President for Research, and American Cancer Society. JNP is a paid consultant for VieCure and Clarified Precision Medicine, and received speaking honoraria from Illumina. SMS received consulting/advisory services/nonpromotional speaking fees from AstraZeneca, Daiichi-Sankyo, Genentech/Roche, Sanofi, Merck, Lilly, Chugai Pharmaceutical Co; research support (to institution) from Genentech/Roche, Kailos Genetics Inc.; serves on a scientific advisory board for Napo Pharmaceuticals; serves on the board of directors for SEAGEN Stock (end 12/14/2023) and Immunome; received third-party writing assistance from Genentech/Roche and AstraZeneca; in-kind travel Seagen, Napo Pharmaceuticals, Sanofi, Daiichi, Sankyo, Genentech/Roche, Chugai Pharmaceutical. JKH is a paid consult for ARUP, Bristol Myers Squibb, and Jackson Laboratories molecular tumor board. All other authors did not report any conflicts of interest.

© 2025 The Author(s). *Clinical Pharmacology & Therapeutics* published by Wiley Periodicals LLC on behalf of American Society for Clinical Pharmacology and Therapeutics.

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial](https://creativecommons.org/licenses/by-nc/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.

1. Grem, J.L. 5-Fluorouracil: forty-plus and still ticking. A review of its preclinical and clinical development. *Investig. New Drugs* **18**, 299–313 (2000).

2. Colon Cancer version 5.2024 (2024) Available from: https://www.nccn.org/professionals/physician_gls/pdf/colon.pdf.
3. Breast Cancer version 6.2024 (2024) Available from: https://www.nccn.org/professionals/physician_gls/pdf/breast.pdf. [cited 2024].
4. Rectal Cancer version 4 (2024) Available from: https://www.nccn.org/professionals/physician_gls/pdf/rectal.pdf. [cited 2024 December 2].
5. Gmeiner, W.H. A narrative review of genetic factors affecting fluoropyrimidine toxicity. *Precis. Cancer Med.* **4**, 38 (2021).
6. Longley, D.B., Harkin, D.P. & Johnston, P.G. 5-Fluorouracil: mechanisms of action and clinical strategies. *Nat. Rev. Cancer* **3**, 330–338 (2003).
7. Meulendijks, D. et al. Clinical relevance of *DPYD* variants c.1679T>G, c.1236G>A/HapB3, and c.1601G>A as predictors of severe fluoropyrimidine-associated toxicity: a systematic review and meta-analysis of individual patient data. *Lancet Oncol.* **16**, 1639–1650 (2015).
8. Meulendijks, D. et al. Patients homozygous for *DPYD* c.1129-5923C>G/haplotype B3 have partial DPD deficiency and require a dose reduction when treated with fluoropyrimidines. *Cancer Chemother. Pharmacol.* **78**, 875–880 (2016).
9. Lee, A.M. et al. *DPYD* variants as predictors of 5-fluorouracil toxicity in adjuvant colon cancer treatment (NCCTG N0147). *J. Natl. Cancer Inst.* **106**, dju298 (2014). <https://doi.org/10.1093/jnci/dju298>.
10. Rosmarin, D. et al. Genetic markers of toxicity from capecitabine and other fluorouracil-based regimens: investigation in the QUASAR2 study, systematic review, and meta-analysis. *J. Clin. Oncol.* **32**, 1031–1039 (2014).
11. Sharma, V., Gupta, S.K. & Verma, M. Dihydropyrimidine dehydrogenase in the metabolism of the anticancer drugs. *Cancer Chemother. Pharmacol.* **84**, 1157–1166 (2019).
12. Diasio, R.B., Beavers, T.L. & Carpenter, J.T. Familial deficiency of dihydropyrimidine dehydrogenase. Biochemical basis for familial pyrimidinemia and severe 5-fluorouracil-induced toxicity. *J. Clin. Invest.* **81**, 47–51 (1988).
13. Diasio, R.B. & Harris, B.E. Clinical pharmacology of 5-fluorouracil. *Clin. Pharmacokinet.* **16**, 215–237 (1989).
14. de With, M. et al. Implementation of dihydropyrimidine dehydrogenase deficiency testing in Europe. *ESMO Open* **8**, 101197 (2023).
15. Hochster, H.S. Routine *DPYD* genetic testing. *J. Clin. Oncol.* **41**, 2119–2120 (2023).
16. Hertz, D.L. Assessment of the clinical utility of pretreatment *DPYD* testing for patients receiving fluoropyrimidine chemotherapy. *J. Clin. Oncol.* **40**, 3882–3892 (2022).
17. Tracksdorf, T. et al. Strategies for *DPYD* testing prior to fluoropyrimidine chemotherapy in the US. *Support Care Cancer* **32**, 497 (2024).
18. El Rouby, N. et al. Global investigation of clinical implementation strategies for *DPYD* and *UGT1A1* pharmacogenetic testing to guide anticancer therapy. *Pharmacogenet. Genom.* **35**, 12–44 (2025).
19. Swen, J.J. et al. A 12-gene pharmacogenetic panel to prevent adverse drug reactions: an open-label, multicentre, controlled, cluster-randomised crossover implementation study. *Lancet* **401**, 347–356 (2023).
20. Damschroder, L.J., Reardon, C.M., Widerquist, M.A.O. & Lowery, J. The updated consolidated framework for implementation research based on user feedback. *Implement. Sci.* **17**, 75 (2022).
21. Horowitz, C.R. et al. The genomic medicine integrative research framework: a conceptual framework for conducting genomic medicine research. *Am. J. Hum. Genet.* **104**, 1088–1096 (2019).
22. Granados, J., Pasternak, A.L., Henry, N.L., Sahai, V. & Hertz, D.L. Risk of toxicity from topical 5-fluorouracil treatment in patients carrying *DPYD* variant alleles. *Clin. Pharmacol. Ther.* **115**, 452–456 (2024).
23. Hicks, J.K. et al. Precision pharmacotherapy: integrating pharmacogenomics into clinical pharmacy practice. *J. Am. Coll. Clin. Pharm.* **2**, 303–313 (2019).

24. Nie, Q. *et al.* Quantitative contribution of rs75017182 to Dihydropyrimidine dehydrogenase mRNA splicing and enzyme activity. *Clin. Pharmacol. Ther.* **102**, 662–670 (2017).
25. Offer, S.M., Wegner, N.J., Fossum, C., Wang, K. & Diasio, R.B. Phenotypic profiling of DPYD variations relevant to 5-fluorouracil sensitivity using real-time cellular analysis and in vitro measurement of enzyme activity. *Cancer Res.* **73**, 1958–1968 (2013).
26. Offer, S.M., Fossum, C.C., Wegner, N.J., Stuflesser, A.J., Butterfield, G.L. & Diasio, R.B. Comparative functional analysis of DPYD variants of potential clinical relevance to dihydropyrimidine dehydrogenase activity. *Cancer Res.* **74**, 2545–2554 (2014).
27. Boisdron-Celle, M. *et al.* 5-fluorouracil-related severe toxicity: a comparison of different methods for the pretherapeutic detection of dihydropyrimidine dehydrogenase deficiency. *Cancer Lett.* **249**, 271–282 (2007).
28. Morel, A. *et al.* Clinical relevance of different dihydropyrimidine dehydrogenase gene single nucleotide polymorphisms on 5-fluorouracil tolerance. *Mol. Cancer Ther.* **5**, 2895–2904 (2006).
29. Sharma, B.B., Rai, K., Blunt, H., Zhao, W., Tosteson, T.D. & Brooks, G.A. Pathogenic DPYD variants and treatment-related mortality in patients receiving fluoropyrimidine chemotherapy: a systematic review and meta-analysis. *Oncologist* **26**, 1008–1016 (2021).
30. Cremolini, C. *et al.* DPYD and UGT1A1 genotyping to predict adverse events during first-line FOLFIRI or FOLFOXIRI plus bevacizumab in metastatic colorectal cancer. *Oncotarget* **9**, 7859–7866 (2018).
31. Hertz, D.L., Smith, D.M., Scott, S.A., Patel, J.N. & Hicks, J.K. Response to the FDA decision regarding DPYD testing prior to fluoropyrimidine chemotherapy. *Clin. Pharmacol. Ther.* **114**, 768–779 (2023).
32. Henricks, L.M. *et al.* Effectiveness and safety of reduced-dose fluoropyrimidine therapy in patients carrying the DPYD*2A variant: a matched pair analysis. *Int. J. Cancer* **144**, 2347–2354 (2019).
33. Deenen, M.J. *et al.* Upfront genotyping of DPYD*2A to individualize fluoropyrimidine therapy: a safety and cost analysis. *J. Clin. Oncol.* **34**, 227–234 (2016).
34. Glewis, S. *et al.* A systematic review and meta-analysis of toxicity and treatment outcomes with pharmacogenetic-guided dosing compared to standard of care BSA-based fluoropyrimidine dosing. *Br. J. Cancer* **127**, 126–136 (2022).
35. Henricks, L.M. *et al.* DPYD genotype-guided dose individualisation of fluoropyrimidine therapy in patients with cancer: a prospective safety analysis. *Lancet Oncol.* **19**, 1459–1467 (2018).
36. Knikman, J.E. *et al.* Survival of patients With cancer With DPYD variant alleles and dose-individualized fluoropyrimidine therapy-a matched-pair analysis. *J. Clin. Oncol.* **41**, 5411–5421 (2023).
37. Amstutz, U. *et al.* Clinical pharmacogenetics implementation consortium (CPIC) guideline for Dihydropyrimidine dehydrogenase genotype and fluoropyrimidine dosing: 2017 update. *Clin. Pharmacol. Ther.* **103**, 210–216 (2018).
38. Lunenburg, C. *et al.* Dutch pharmacogenetics working group (DPWG) guideline for the gene-drug interaction of DPYD and fluoropyrimidines. *Eur. J. Hum. Genet.* **28**, 508–517 (2020).
39. CPIC® Guideline for Fluoropyrimidines and DPYD (2024) Available from: <https://cpicpgx.org/guidelines/guideline-for-fluoropyrimidines-and-dpyd/>. [cited 2024 December 2].
40. Capecitabine Prescribing Info Available from: <https://www.pharmgkb.org/chemical/PA448771/prescribingInfo>. [cited 2024].
41. Head and Neck Cancers version 1.2025 (2024) Available from: https://www.nccn.org/professionals/physician_gls/pdf/head-and-neck.pdf. [cited 2024 December 2].
42. Henricks, L.M., Opdam, F.L., Beijnen, J.H., Cats, A. & Schellens, J.H.M. DPYD genotype-guided dose individualization to improve patient safety of fluoropyrimidine therapy: call for a drug label update. *Ann. Oncol.* **28**, 2915–2922 (2017).
43. Morris, V.K. *et al.* Treatment of metastatic colorectal cancer: ASCO guideline. *J. Clin. Oncol.* **41**, 678–700 (2023).
44. Baxter, N.N. *et al.* Adjuvant therapy for stage II colon cancer: ASCO guideline update. *J. Clin. Oncol.* **40**, 892–910 (2022).
45. Khorana, A.A. *et al.* Potentially curable pancreatic adenocarcinoma: ASCO clinical practice guideline update. *J. Clin. Oncol.* **37**, 2082–2088 (2019).
46. Denduluri, N. *et al.* Selection of optimal adjuvant chemotherapy and targeted therapy for early breast cancer: ASCO guideline update. *J. Clin. Oncol.* **39**, 685–693 (2021).
47. Advocates for Universal DPD/DPYD Testing (AUDT): Initiatives Available from: <https://test4dpd.org/initiatives/>. [cited 2024].
48. Baker, S.D. *et al.* DPYD testing: time to put patient safety first. *J. Clin. Oncol.* **41**, 2701–2705 (2023).
49. Citizen Petition from Kenneth E. Surprenant. 2022 (2020).
50. Fluorouracil [package insert] (2024).
51. Capecitabine [package insert] (2022).
52. Capecitabine Drug Label Annotations Available from: <https://www.pharmgkb.org/chemical/PA448771/labelAnnotation>. [cited 2024].
53. Nguyen, D.G. *et al.* Real-world impact of an in-house Dihydropyrimidine dehydrogenase (DPYD) genotype test on fluoropyrimidine dosing, toxicities, and hospitalizations at a multisite cancer center. *JCO Precis. Oncol.* **8**, e2300623 (2024).
54. Empey, P.E., Pratt, V.M., Hoffman, J.M., Caudle, K.E. & Klein, T.E. Expanding evidence leads to new pharmacogenomics payer coverage. *Genet. Med.* **23**, 830–832 (2021).
55. Mason, N.T., Bell, G.C., Quilitz, R.E., Greene, J.N. & McLeod, H.L. Budget impact analysis of CYP2C19-guided voriconazole prophylaxis in AML. *J. Antimicrob. Chemother.* **70**, 3124–3126 (2015).
56. Morris, S.A. *et al.* Addressing barriers to increased adoption of DPYD genotyping at a large multisite cancer center. *Am. J. Health Syst. Pharm.* **80**, 1342–1349 (2023).
57. Varughese, L.A. *et al.* DPYD and UGT1A1 pharmacogenetic testing in patients with gastrointestinal malignancies: an overview of the evidence and considerations for clinical implementation. *Pharmacotherapy* **40**, 1108–1129 (2020).
58. Innocenti, F., Mills, S.C., Sanoff, H., Ciccolini, J., Lenz, H.J. & Milano, G. All you need to know about DPYD genetic testing for patients treated with fluorouracil and capecitabine: a practitioner-friendly guide. *JCO Oncol. Pract.* **16**, 793–798 (2020).
59. Toffoli, G. *et al.* The genotype for DPYD risk variants in patients With colorectal cancer and the related toxicity management costs in clinical practice. *Clin. Pharmacol. Ther.* **105**, 994–1002 (2019).
60. Brooks, G.A., Tapp, S., Daly, A.T., Busam, J.A. & Tosteson, A.N.A. Cost-effectiveness of DPYD genotyping prior to fluoropyrimidine-based adjuvant chemotherapy for colon cancer. *Clin. Colorectal Cancer* **21**, e189–e195 (2022).
61. Ontario, H. DPYD genotyping in patients who have planned cancer treatment With fluoropyrimidines: a health technology assessment. *Ont. Health Technol. Assess. Ser.* **21**, 1–186 (2021).
62. Henricks, L.M. *et al.* A cost analysis of upfront DPYD genotype-guided dose individualisation in fluoropyrimidine-based anticancer therapy. *Eur. J. Cancer* **107**, 60–67 (2019).
63. Fragoulakis, V. *et al.* Cost-utility analysis and cross-country comparison of pharmacogenomics-guided treatment in colorectal cancer patients participating in the U-PGx PREPARE study. *Pharmacol. Res.* **197**, 106949 (2023).
64. Murphy, C. *et al.* Cost implications of reactive versus prospective testing for Dihydropyrimidine dehydrogenase deficiency in patients With colorectal cancer: a single-institution experience. *Dose-Response* **16**, 1559325818803042 (2018).
65. Lau-Min, K.S. *et al.* Preemptive pharmacogenetic testing to guide chemotherapy dosing in patients with gastrointestinal malignancies: a qualitative study of barriers to implementation. *BMC Cancer* **22**, 47 (2022).
66. Varughese, L.A. *et al.* Implementing pharmacogenetic testing in gastrointestinal cancers (IMPACT-GI): study protocol for a pragmatic implementation trial for establishing DPYD and UGT1A1 screening to guide chemotherapy dosing. *Front. Oncol.* **12**, 859846 (2022).

67. Jacobson, J.O. *et al.* Design and implementation of an opt-out, end-to-end, preemptive DPYD testing program for patients planned for a systemic fluoropyrimidine. *JCO Oncol. Pract.* **20**, 1115–1122 (2024).
68. Thompson, J.T., Wood, D.M. & Dargan, P.I. Review of the fluoropyrimidine antidote uridine triacetate. *Br. J. Clin. Pharmacol.* (2024). doi: [10.1111/bcp.16319](https://doi.org/10.1111/bcp.16319). Epub ahead of print.
69. Leung, M., Rogers, J.E. & Shureiqi, I. Use of uridine triacetate to reverse severe persistent myelosuppression following 5-fluorouracil exposure in a patient With a c.557A>G heterozygous DPYD variant. *Clin. Colorectal Cancer* **20**, 273–278 (2021).
70. Pratt, V.M. *et al.* DPYD genotyping recommendations: a joint consensus recommendation of the Association for Molecular Pathology, American College of Medical Genetics and Genomics, clinical pharmacogenetics implementation consortium, College of American Pathologists, Dutch pharmacogenetics working Group of the Royal Dutch Pharmacists Association, European Society for Pharmacogenomics and Personalized Therapy, pharmacogenomics knowledgebase, and Pharmacogene variation consortium. *J. Mol. Diagn.* **26**, 851–863 (2024).
71. Vo, T.T., Bell, G.C., Owusu Obeng, A., Hicks, J.K. & Dunnenberger, H.M. Pharmacogenomics implementation: considerations for selecting a reference laboratory. *Pharmacotherapy* **37**, 1014–1022 (2017).
72. Bousman, C.A., Zierhut, H. & Muller, D.J. Navigating the labyrinth of pharmacogenetic testing: a guide to test selection. *Clin. Pharmacol. Ther.* **106**, 309–312 (2019).
73. Seligson, N.D. *et al.* Integrating pharmacogenomic testing into paired germline and somatic genomic testing in patients with cancer. *Pharmacogenomics* **24**, 731–738 (2023).
74. Tremmel, R., Hübschmann, D., Schaeffeler, E., Pirmann, S., Fröhling, S. & Schwab, M. Innovation in cancer pharmacotherapy through integrative consideration of germline and tumor genomes. *Pharmacol. Rev.* **77**, 100014 (2024).
75. Sutton, R.T., Pincock, D., Baumgart, D.C., Sadowski, D.C., Fedorak, R.N. & Kroeker, K.I. An overview of clinical decision support systems: benefits, risks, and strategies for success. *NPJ Digit. Med.* **3**, 17 (2020).
76. Hearst, E.C. *et al.* The future is now: implementation of standard of care, preemptive pharmacogenomic testing in patients with gastrointestinal cancers. *J. Clin. Oncol.* **42**, 1546 (2024).
77. Wake, D.T., Smith, D.M., Kazi, S. & Dunnenberger, H.M. Pharmacogenomic clinical decision support: a review, how-to guide, and future vision. *Clin. Pharmacol. Ther.* **112**, 44–57 (2022).
78. Smith, D.M., Wake, D.T. & Dunnenberger, H.M. Pharmacogenomic clinical decision support: a scoping review. *Clin. Pharmacol. Ther.* **113**, 803–815 (2023).
79. Muldoon, M. *et al.* Real-world implementation of and pharmacogenetic testing in a community-based cancer center. *Clin. Transl. Sci.* **17**, e13704 (2024).
80. Allen, A.J. *et al.* A precision medicine navigator can mitigate inequities associated with utilization of genomic tests in black men with prostate cancer. *Int. J. Radiat. Oncol. Biol. Phys.* **117**, S15–S16 (2023).
81. Hicks, J.K. *et al.* Prospective CYP2C19-guided voriconazole prophylaxis in patients With neutropenic acute myeloid leukemia reduces the incidence of subtherapeutic antifungal plasma concentrations. *Clin. Pharmacol. Ther.* **107**, 563–570 (2020).
82. Hicks, J.K. *et al.* Integrating somatic and germline next-generation sequencing into routine clinical oncology practice. *JCO Precis. Oncol.* **5**, 895 (2021).
83. Morris, S.A., Nguyen, D.G., Morris, V., Mroz, K., Kwange, S.O. & Patel, J.N. Integrating pharmacogenomic results in the electronic health record to facilitate precision medicine at a large multisite health system. *J. Am. College Clin. Pharm.* **7**, 845–857 (2024).
84. Shugg, T. *et al.* Development of a multifaceted program for pharmacogenetics adoption at an Academic Medical Center: practical considerations and lessons learned. *Clin. Pharmacol. Ther.* **116**, 914–931 (2024).
85. Claudio-Campos, K. *et al.* Acceptability, feasibility, and utility of integrating pharmacogenetic testing into a child psychiatry clinic. *Clin. Transl. Sci.* **14**, 589–598 (2021).
86. Adams, S.M. *et al.* Advancing pharmacogenomics education in the Core PharmD curriculum through student personal genomic testing. *Am. J. Pharm. Educ.* **80**, 3 (2016).
87. Wake, D.T., Bell, G.C., Gregornik, D.B., Ho, T.T. & Dunnenberger, H.M. Synthesis of major pharmacogenomics pretest counseling themes: a multisite comparison. *Pharmacogenomics* **22**, 165–176 (2021).
88. Ho, T.T., Bell, G., Gammal, R.S., Gregornik, D., Wake, D.T. & Dunnenberger, H.M. A clinician's guide for counseling patients on results of a multigene pharmacogenomic panel. *Am. J. Health Syst. Pharm.* **79**, 1634–1644 (2022).
89. Ho, T.T. *et al.* Integrating DPYD/UGT1A1 with hereditary cancer germline genetic education via a point-of-care digital education tool for patients with pancreatic cancer. *Pharmacogenet. Genom.* **35**, 12–44 (2025).
90. Lam, S.W., Guchelaar, H.J. & Boven, E. The role of pharmacogenetics in capecitabine efficacy and toxicity. *Cancer Treat. Rev.* **50**, 9–22 (2016).
91. Turner, A.J. *et al.* Updated DPYD HapB3 haplotype structure and implications for pharmacogenomic testing. *Clin. Transl. Sci.* **17**, e13699 (2024).
92. Bembenek, B.M., Joshy, A. & Offer, S.M. Lethal 5-fluorouracil toxicity in a carrier of DPYD c.704G>A (p.R235Q). *JCO Precis. Oncol.* **8**, e2400294 (2024).
93. Wu, A. *et al.* Implementation of pharmacogenetic testing in oncology: DPYD-guided dosing to prevent fluoropyrimidine toxicity in British Columbia. *Front. Pharmacol.* **14**, 1257745 (2023).
94. Kleinjan, J.P., Brinkman, I., Bakema, R., van Zanden, J.J. & van Rooijen, J.M. Tolerance-based capecitabine dose escalation after DPYD genotype-guided dosing in heterozygote DPYD variant carriers: a single-center observational study. *Anti-Cancer Drugs* **30**, 410–415 (2019).
95. Abdel-Rahman, O. & Karachiwala, H. Impact of age on toxicity and efficacy of 5-FU-based combination chemotherapy among patients with metastatic colorectal cancer; a pooled analysis of five randomized trials. *Int. J. Color. Dis.* **34**, 1741–1747 (2019).
96. Milano, G. *et al.* Influence of sex and age on fluorouracil clearance. *J. Clin. Oncol.* **10**, 1171–1175 (1992).
97. Meulendijks, D., Cats, A., Beijnen, J.H. & Schellens, J.H.M. Improving safety of fluoropyrimidine chemotherapy by individualizing treatment based on dihydropyrimidine dehydrogenase activity – ready for clinical practice? *Cancer Treat. Rev.* **50**, 23–34 (2016).
98. Knikman, J.E., Gelderblom, H., Beijnen, J.H., Cats, A., Guchelaar, H.J. & Henricks, L.M. Individualized dosing of fluoropyrimidine-based chemotherapy to prevent severe fluoropyrimidine-related toxicity: what are the options? *Clin. Pharmacol. Ther.* **109**, 591–604 (2021).
99. Chao, C.J. *et al.* Administration mode matters for 5-fluorouracil therapy: Physiologically based pharmacokinetic evidence for avoidance of myelotoxicity by continuous infusion but not intravenous bolus. *Br. J. Clin. Pharmacol.* (2024). doi: [10.1111/bcp.16061](https://doi.org/10.1111/bcp.16061). Epub ahead of print.
100. de With, M. *et al.* Dihydropyrimidine dehydrogenase phenotyping using pretreatment uracil: a note of caution based on a large prospective clinical study. *Clin. Pharmacol. Ther.* **112**, 62–68 (2022).
101. Thomas, F. *et al.* Plasma uracil as a DPD phenotyping test: pre-analytical handling matters! *Clin. Pharmacol. Ther.* **113**, 471–472 (2023).